

Where TTS Fits in Voice AI Pipeline



The voice AI pipeline transforms audio input into audio output through several stages: Audio → ASR → Text → LLM → Text → TTS → Audio.

TTS is the final stage that converts AI responses into human-understandable speech. Without TTS, voice assistants would remain silent, unable to communicate their responses.

What is Text-to-Speech (TTS)?



Text-to-Speech is the process of converting written text into spoken audio, enabling machines to communicate verbally with humans.

Real-World Examples of TTS



Voice Assistants

Alexa, Google Assistant, and Siri use TTS to respond to user queries



Navigation Systems

Google Maps and car navigation provide turn-by-turn voice directions



Accessibility Tools

Screen readers help visually impaired users access digital content



Customer Service

Voice bots handle enquiries with automated spoken responses

Why TTS is Important in Voice AI

Human-Machine Communication

Enables natural dialogue between people and AI systems

Hands-Free Interaction

Allows users to multitask whilst receiving information

Accessibility

Provides visually impaired users access to digital content

Voice improves user experience compared to text alone. It makes AI interactions feel more natural and human-like.



Challenges in Text-to-Speech

TTS is far more complex than simply reading text aloud. Creating natural-sounding speech requires solving multiple technical challenges simultaneously.

Poor TTS implementation results in robotic, unnatural voices that diminish user experience and reduce engagement with voice AI systems.

Pronunciation Correctness

Accurately pronouncing words, names, and technical terms

Natural Tone

Conveying appropriate emotion and emphasis

Correct Pauses

Placing natural breaks between words and phrases

Natural Rhythm

Maintaining human-like speech cadence and flow

Stage 1: Text Processing



Converting Text to Phonemes

Text is converted into phonemes—the basic sound units of language. This process helps the model pronounce words correctly.

📌 **Example:** The word "cat" is broken down into phonemes: /k/ /æ/ /t/

Phonetic representation ensures consistent pronunciation across different words and contexts.

Stage 2: Acoustic Model



Phoneme Input

Receives phonetic representation from text processing



Feature Generation

Converts phonemes into acoustic features representing pitch, tone, and duration



Speech Definition

Defines how the speech will sound before audio generation

The acoustic model bridges the gap between text representation and audio generation, defining the characteristics of the speech.

Stage 3: Vocoder

The vocoder converts acoustic features into an audio waveform, generating the actual sound signal. This is the final step that produces the audio output listeners hear.

Modern neural vocoders have dramatically improved the quality and naturalness of synthesised speech.



TRADITIONAL APPROACH

Traditional TTS Systems

Rule-Based and Concatenative Methods

Traditional TTS systems use pre-recorded speech segments that are stitched together to form complete sentences.

Whilst this approach provided clear pronunciation, it suffered from significant limitations that made it unsuitable for modern voice AI applications.

Concatenative TTS

How It Works

Concatenative TTS joins pre-recorded audio segments to form speech output.

Advantages

- Clear pronunciation
- Predictable output
- Simple implementation

Disadvantages

- Sounds unnatural
- Cannot generate new voice styles
- Limited expressiveness

MODERN APPROACH

Neural TTS Systems



Deep Learning Models

Neural TTS uses deep learning to learn speech patterns from data



Tacotron

End-to-end model that generates mel spectrograms from text



WaveNet

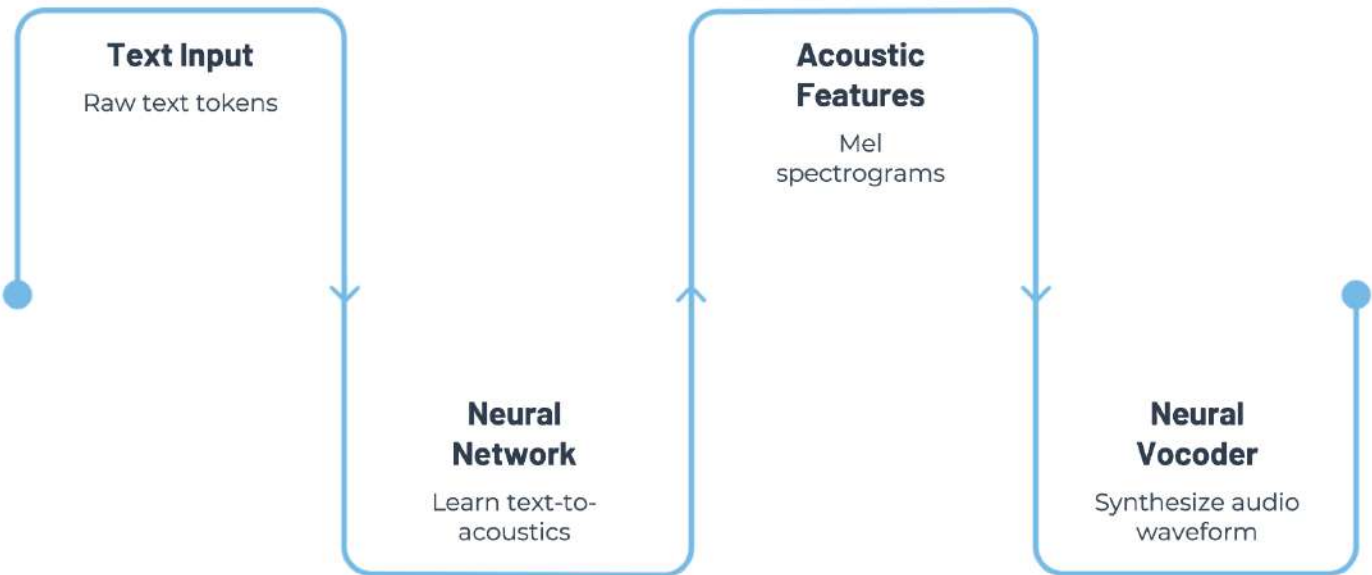
Generates raw audio waveforms with exceptional quality

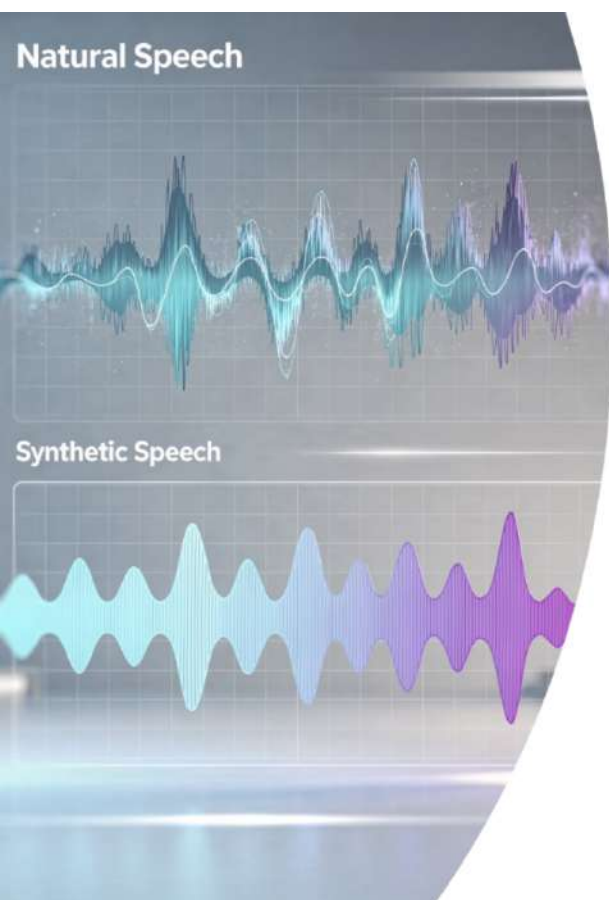


FastSpeech

Parallel generation for faster, more efficient synthesis

Neural TTS Pipeline





Advantages of Neural TTS



More Natural Speech

Produces human-like prosody and intonation



Better Pronunciation

Handles complex words and names accurately



Multiple Voices

Supports diverse voice characteristics and styles



Emotion and Tone

Conveys appropriate emotional expression

Neural TTS has become the industry standard due to these significant advantages over traditional methods.

Neural Vocoders

Generating Realistic Speech Waveforms

Neural vocoders are specialised models that transform acoustic features into high-quality audio waveforms.

These models have revolutionised TTS by producing natural-sounding audio that closely resembles human speech.



WaveNet

Autoregressive model generating samples sequentially



HiFi-GAN

Fast, high-fidelity generative adversarial network



WaveGlow

Flow-based model for parallel waveform generation

Voice Characteristics in TTS



Pitch

Controls high or low tonal quality of the voice, affecting perceived gender and age



Speed

Determines fast or slow speaking rate, impacting comprehension and urgency



Emotion

Conveys feelings like happiness, sadness, excitement, or neutral delivery



Accent

Represents regional pronunciation patterns and cultural voice characteristics

Neural TTS systems can precisely control these voice attributes, enabling customised and contextually appropriate speech output.



TTS Latency

Latency is the time required to generate speech from text input. Low latency is critical for real-time applications where conversational flow must feel natural.

High latency causes noticeable delays in conversation, breaking the illusion of natural interaction. Voice assistants require minimal latency to maintain user engagement.

Factors Affecting TTS Latency

Optimising for Speed

Smaller, more efficient models generate speech faster, making them suitable for real-time applications.

Hardware acceleration with GPUs significantly reduces processing time. Streaming generation allows audio playback to begin before synthesis completes.

Streaming vs Batch TTS

Streaming TTS

Generates speech incrementally whilst text is still arriving, enabling low-latency responses. Ideal for voice assistants and real-time applications.

- Lower perceived latency
- Faster initial response
- More complex implementation

Batch TTS

Generates speech only after receiving complete text input, resulting in higher latency. Suitable for non-interactive applications like audiobook generation.

- Higher latency
- Better audio consistency
- Simpler implementation

QUALITY ASSESSMENT

Evaluating TTS Quality



High-quality TTS achieves human-like naturalness whilst maintaining perfect intelligibility and minimal latency.

Mean Opinion Score (MOS)

Measuring Perceived Audio Quality

MOS is the standard metric for evaluating TTS quality. Human listeners rate synthesised speech on a scale from 1 to 5.

This subjective measure captures the overall listening experience, including naturalness, clarity, and pleasantness of the voice.

